i. How does blockchain invalidate double spending?

- Blockchain uses consensus mechanisms like Proof-of-Work (PoW) or Proof-of-Stake (PoS) to validate and agree on the order of transactions. Once a transaction is confirmed in a block and added to the blockchain, attempting to spend the same funds again would require altering multiple subsequent blocks, making it computationally infeasible and easily detectable.

ii. How is PoW Byzantine fault-tolerant? (CO1)

- Proof-of-Work (PoW) in blockchain ensures Byzantine fault tolerance by making it computationally expensive for malicious nodes to control the consensus. Nodes must solve complex mathematical puzzles to propose and validate blocks, requiring honest participants to invest computational power. Byzantine fault tolerance is achieved as long as the majority of participants are honest.

"Increased hash rate means the blockchain is more secured." - Explain. (POI)

- A higher hash rate in a blockchain network indicates more computational power is devoted to securing the network through processes like mining. This increased computational effort makes it more difficult for malicious actors to control the network, enhancing security against attacks like 51% attacks.

jv. What hashing algorithm is used in Bitcoin? (CO1)

- Bitcoin primarily uses the SHA-256 (Secure Hash Algorithm 256-bit) hashing algorithm.

V. Write a simple hashing technique.

- A simple hashing technique involves adding up the ASCII values of characters in a string and taking the remainder when divided by a chosen prime number.